AGRICULTURAL CROP PRODUCTION DASHBOARD USING TABLEAU

1. INTRODUCTION

1.1 Project Overview

The Agricultural Crop Production Dashboard is an interactive Tableau-based visualization project that analyzes crop production data across different states, districts, seasons, and crops. It provides meaningful insights through charts, maps, and dashboards to support data-driven agricultural decisions.

1.2 Purpose

The purpose of this project is to simplify agricultural data analysis by presenting crop production information in an interactive dashboard. It helps users identify production trends, compare regions, and improve agricultural planning.

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2. IDEATION PHASE

2.1 Problem Statement

Agricultural crop production data is large, scattered, and difficult to analyze using traditional methods. This makes it challenging for farmers, researchers, and policymakers to make informed decisions.

2.2 Empathy Map Canvas

Farmers, researchers, and policymakers need an easy way to analyze crop production data. They face difficulties due to large datasets and manual analysis. An interactive dashboard helps them gain quick insights for better decision-making.

2.3 Brainstorming

Several ideas were explored, including weather analysis, crop recommendation, and market price analysis. The Agricultural Crop Production Dashboard was selected because it provides interactive visualizations, supports easy comparison, and improves agricultural planning.

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3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Users access the dashboard, apply filters for state, crop, district, and season, analyze charts and maps, identify production trends, and use insights for better planning and decision-making.

3.2 Solution Requirement

Functional Requirements

- Interactive Dashboard
- Crop-wise Analysis
- State-wise Analysis
- Season-wise Analysis
- District-wise Analysis
- Story Visualization

Non-Functional Requirements

- User-friendly interface
- High performance
- Reliable results
- Secure data handling
- Scalability
- Easy availability

3.3 Data Flow Diagram

Dataset → Data Cleaning → Tableau Processing → Dashboard → Story → User Insights

3.4 Technology Stack

- Tableau Desktop
- Microsoft Excel (Dataset)
- Windows Operating System
- GitHub (Project Repository)

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4. PROJECT DESIGN

4.1 Problem Solution Fit

The dashboard solves the problem of complex agricultural data analysis by providing interactive visualizations and filters that help users understand production trends quickly.

4.2 Proposed Solution

Develop an interactive Tableau dashboard that enables users to analyze agricultural crop production using charts, maps, filters, and stories for better decision-making.

4.3 Solution Architecture

Agricultural Dataset
↓
Data Preprocessing
↓
Tableau Analysis
↓
Interactive Dashboard
↓
Story Visualization
↓
Decision Making

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5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint 1

- Data Collection
- Data Cleaning
- Data Preparation

Sprint 2

- Data Visualization
- Dashboard Development
- Story Creation

Velocity = **16 Story Points per Sprint**

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6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

- Dataset successfully loaded
- Missing values handled
- Interactive filters implemented
- Calculated fields created
- Dashboard contains **6 visualizations**
- Story contains **6 visualizations**

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7. RESULTS

7.1 Output Screenshots

Include screenshots of:

- Dataset
- Tableau Dashboard
- Story
- Filters
- Charts
- Maps

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8. ADVANTAGES & DISADVANTAGES

Advantages

- Interactive dashboard
- Easy data visualization
- Quick comparison of crops and states
- Supports better decision-making
- Time-saving analysis

Disadvantages

- Depends on dataset quality
- Requires Tableau Desktop
- No real-time data updates

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9. CONCLUSION

The Agricultural Crop Production Dashboard successfully transforms complex agricultural data into meaningful visual insights. It enables users to analyze crop production efficiently and supports informed agricultural planning through interactive dashboards and stories.

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10. FUTURE SCOPE

- Real-time agricultural data integration

- Weather data analysis
- Market price prediction
- AI-based crop recommendations
- Mobile dashboard support